

CONTROL

QWEN2.5-7B-INSTRUCT

— Boundary token $\rightarrow$ next sentence, R^2 (WikiText, mean of 4 tokens, 1,200 pairs)

PREDICTOR

METRIC

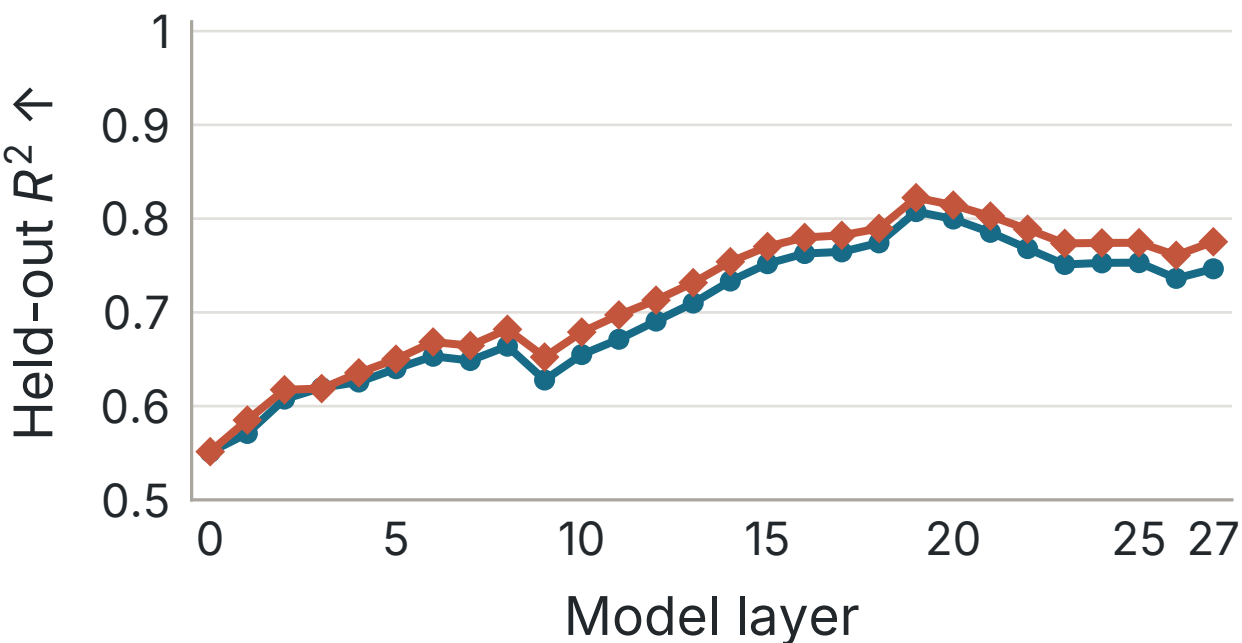
—●— Linear —◆— Nonlinear

A · 50,000 TRAINING CONTEXTS

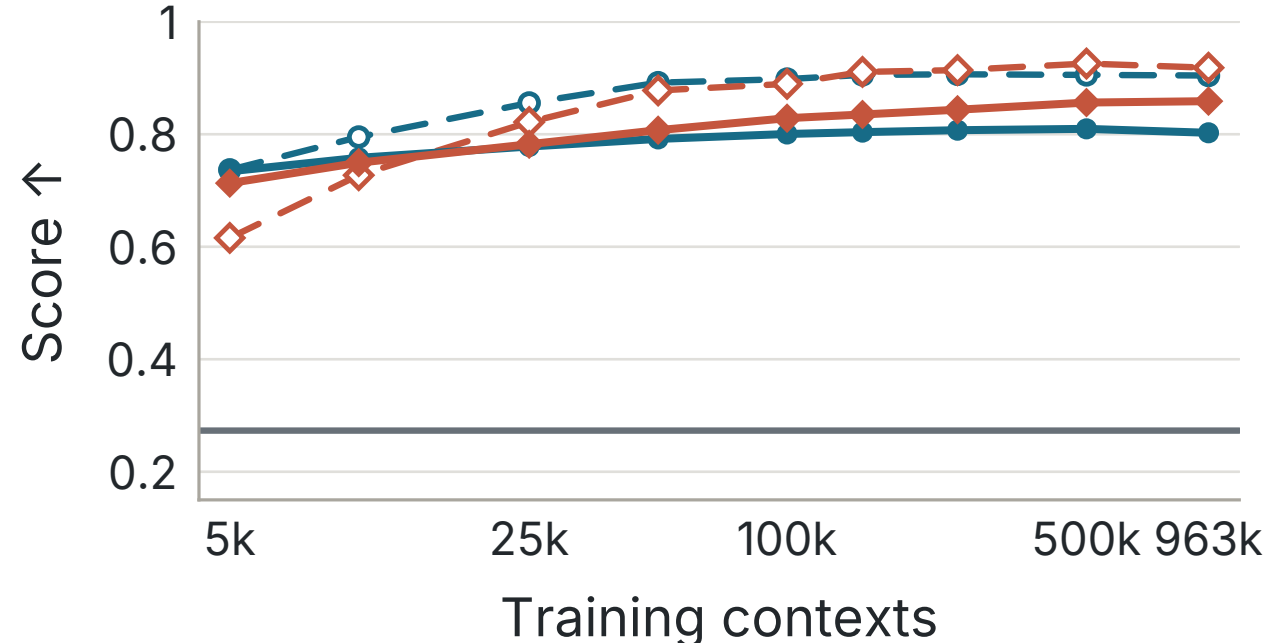
—●— Held-out R^2 —○— Top-1 retrieval

B · LAYER 19, 10,000-CANDIDATE RETRIEVAL

Predictability across layers



Scaling with training data



C · LAYER 19, 25,000 CONTEXTS, 10,000 CANDIDATES

Baselines

Top-1 retrieval $\rightarrow$ Held-out $R^2 \rightarrow$

Linear map

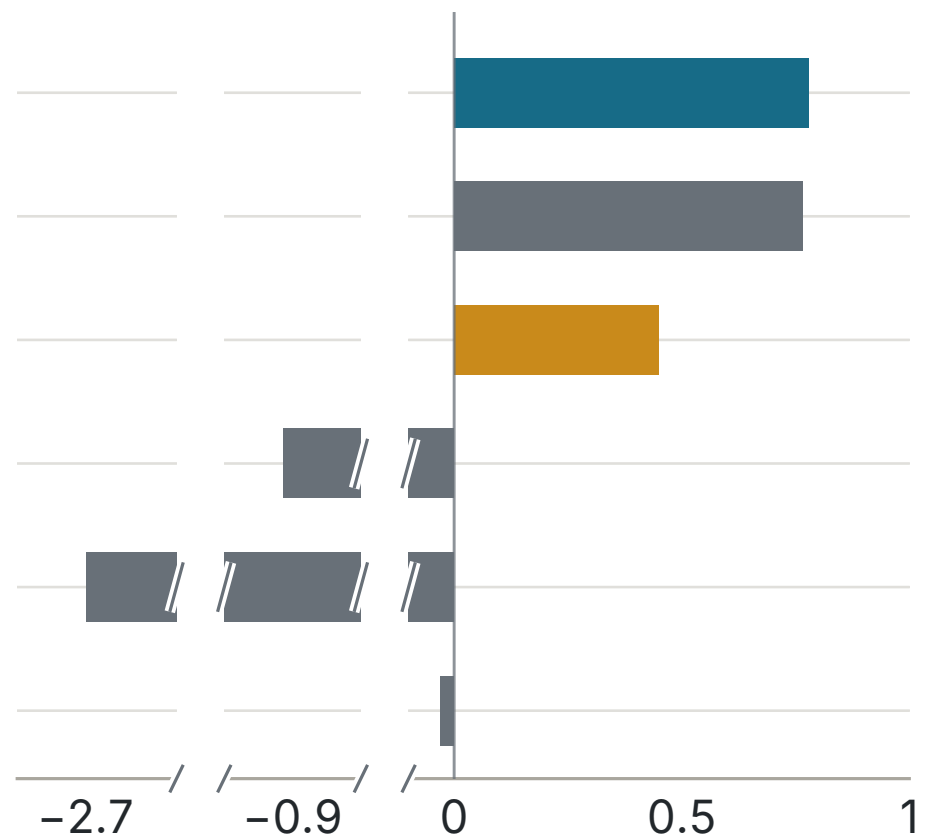
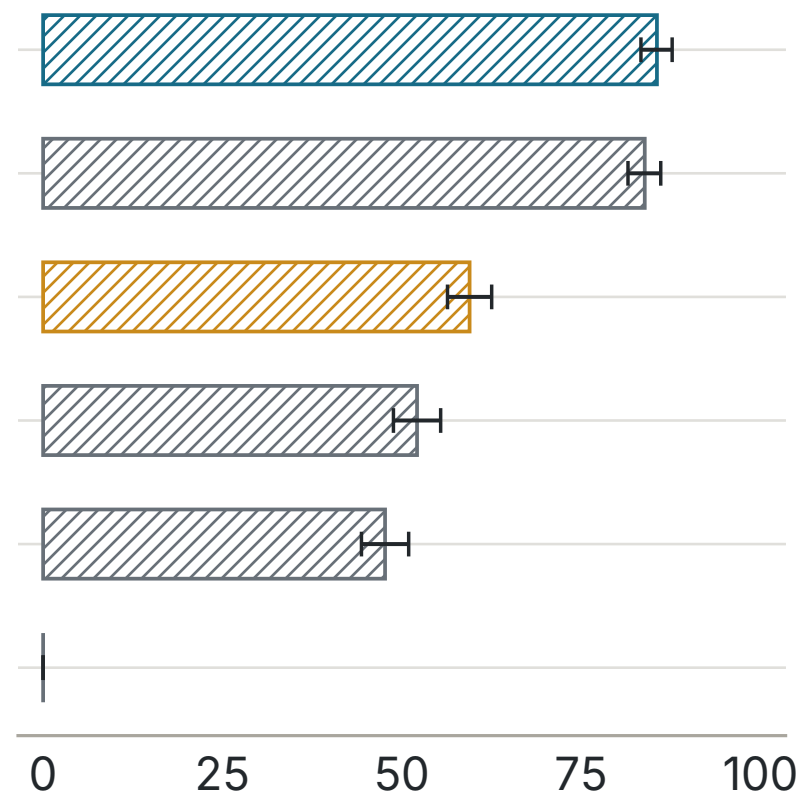
PCA-1024

Encoder (BGE)

Copy + bias

Copy

Shuffled pairs



Accuracy (%)